PROKON	Job Number Prokon	Sheet 0
Your details here	Job Title Prokon	
	Client Aathi	
	Calcs by / Checked by / Date	

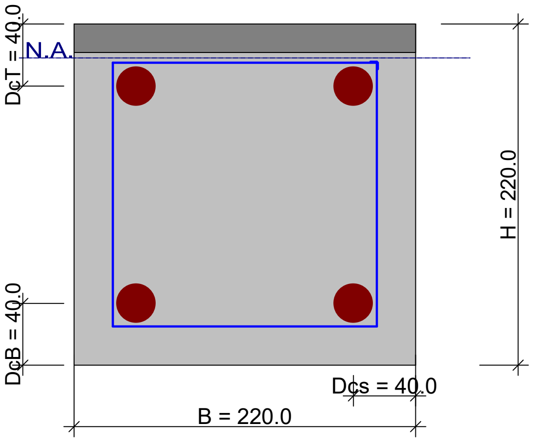
Concrete Section design :

C22

Input Data

ULS Bending Moment M	(kNm)	15.79
ULS Torsion Moment T	(kNm)	0
ULS Shear Force V	(kN)	8.308
Web width B	(mm)	220
Total height H	(mm)	220
Flange Width Bf	(mm)	
Flange Height Hf	(mm)	
Reinf centroid depth DcT	(mm)	40
Reinf centroid depth DcB	(mm)	40
Reinf depth sides DcS	(mm)	40
fc'	(MPa)	30
fy - Main bars	(MPa)	460
fyv - Links	(MPa)	250
% Redistribution		0

ACI 318 - 2014



N.A. depth = 21.9 mm

Output

Design Results										
Moment			Shear			Torsion (entire section)				
Muc	59.1	kNm	Vc	26.30	kN	T	60.00	kNm		
As	223	mm^2	Vs	0.00	kN	Tc	10.98	kNm		
As'	0	mm^2	Asv/Sv	0.0000		Asv/Sv	0.0000			
Amin	121	mm^2	Asv/Sv nom	0.3080		As	0	mm^2		
Suggested Reinforcement Configurations										
Bars (As)	mm^2		Bars	(Asv/sv)		Bars	(Asv/sv)			
5Y8	251		2 8@300	0.34						
3Y10	236		2 10@500	0.31						
2Y12	226		2 12@500	0.45						
Bars (As)	mm^2					Longitudinal Bars			Longitudinal Bars	
Combined Longitudinal bars for Moment and Web Torsion										
Bottom Bars			Top Bars							
Bars	mm^2		Bars	mm^2						
Maximum allowable moment redistribution for this section is20.00 % - ACI318: 6.6.5										

DETAILED CALCULATIONS

Design code :ACI 318 - 2014
Country:UK

Flexural capacity and reinforcement calculations:

Maximum concrete strain 22.2.2.1

$\varepsilon_c = 0.003$

Maximum tensile stress in steel 21.2.2.2

$f_{st} = f_y = 460.000$ MPa

Maximum tensile stress in steel 21.2.2.2

$f_{sd} = f_y = 460.000$ MPa

Equivalent rectangular stress block parameters:

22.2.2.4.3

$$\beta = 0.85 - \frac{0.05 \cdot (f_{c'} - 28)}{7}$$

$$= 0.85 - \frac{0.05 \times (30 - 28)}{7}$$

$$= 0.8357$$

22.2.2.4.1

$\alpha = 0.85$

21.2.1

$\phi = 0.90$

Calculation of the maximum steel strain:

$$\varepsilon_{st} = \frac{f_{st}}{200\times10^3}$$

$$= \frac{460}{200\times10^3}$$

$$= 0.0023$$

Balanced neutral axis

$$x_b = \frac{d}{1 + \frac{\varepsilon_{st}}{\varepsilon_c}}$$

$$= \frac{180}{1 + \frac{.0023}{.003}}$$

$$= 101.887 \text{ mm}$$

Maximum allowed neutral axis depth ratio for moment redistribution: (beam) 6.6.5

$$\rho_{max} = \frac{\frac{\alpha\cdot \beta\cdot f_{c'}\cdot c_{strain}}{c_{strain} + 0.0075}}{f_y}$$

$$= \frac{\frac{.85\times.83572\times30\times.003}{.003 + 0.0075}}{460}$$

$$= 0.0132$$

Depth of rectangular stress block:

$$a = \beta\cdot x$$

$$= .83572\times101.89$$

$$= 85.152 \text{ mm}$$

Concrete moment capacity: 21.2.2.2

$$M_{rc} = a\cdot \left[d - \frac{a}{2}\right]\cdot b\cdot \alpha\cdot \phi\cdot f_{c'}\cdot 1\times10^{-6}$$

$$= 85.149\times\left[180 - \frac{85.149}{2}\right]\times220\times.85\times.9\times30\times1\times10^{-6}$$

$$= 59.082 \text{ kNm}$$

Lever arm 21.2.2.2

$$z = d\cdot\left[0.5 + \sqrt{0.25 - \frac{\frac{M}{\phi\cdot b\cdot d^2\cdot f_{c'}}}{2\cdot\alpha}}\right]$$

$$= 180\times\left[0.5 + \sqrt{0.25 - \frac{\frac{1579\times10^4}{.9\times220\times180^2\times30}}{2\times.85}}\right]$$

$$= 170.848 \text{ mm}$$

Tension steel 21.2.2.2

$$A_{st} = \frac{M}{\phi\cdot f_{st}\cdot z}$$

$$= \frac{1579\times10^4}{.9\times460\times170.85}$$

$$= 223.237 \text{ mm}^2$$

Neutral axis depth 21.2.2.2

$$N_A = \frac{2\cdot(d - z)}{\beta}$$

$$= \frac{2\times(180 - 170.85)}{.83572}$$

$$= 21.897 \text{ mm}$$

Minimum Flexural Reinforcement

9.6.1.2

$$A_{min} = \frac{0.25\cdot\sqrt{f_{c'}}\cdot b_w\cdot d}{f_y}$$

$$= \frac{0.25\times\sqrt{30}\times220\times180}{460}$$

$$= 117.879 \text{ mm}^2$$

9.6.1.2

$$A_{min2} = \frac{1.4 \cdot b_w \cdot d}{f_y}$$
$$= \frac{1.4 \times 220 \times 180}{460}$$
$$= 120.522 \text{ mm}^2$$

$$A_{min} = \max(A_{min}, A_{min2})$$
$$= \max(, 120.52)$$
$$= 120.520$$

Shear Calculations:

Area used for shear calculations:
Shear area 22.5.5

$$A_{sh} = b_w \cdot d_e$$
$$= 220 \times 180$$
$$= 39.60 \times 10^3 \text{ mm}^2$$

Reinforcement ratio 22.5.5

$$F_1 = \frac{A_{st}}{A_{sh}}$$
$$= \frac{223.24}{39600}$$
$$= 0.0056$$

22.5.5

$$v_1 = \sqrt{f_{c'}} = 5.477 \text{ MPa}$$

$$v_1 = \min(v_1, 8.3)$$
$$= \min(, 8.3)$$
$$= 0.0000 \times 10^0$$

$$F_2 = \frac{V \cdot d_e}{M}$$

$$= \frac{8.308 \times .18}{15.79}$$

$$= 0.0947$$

$$f_2 = \min(f_2, 1.0)$$

$$= \min(, 1.0)$$

$$= 0.0000 \times 10^0$$

22.5.5

$$v_c = 0.75 \cdot (0.16 \cdot v_1 + 17 \cdot F_1 \cdot F_2)$$

$$= 0.75 \times (0.16 \times 5.4772 + 17 \times .00564 \times .09471)$$

$$= 0.6641 \text{ MPa}$$

Shear force capacity 22.5.5

$$V_{co} = v_c \cdot b_w \cdot d_e \cdot 1 \times 10^{-3}$$

$$= .66407 \times 220 \times 180 \times 1 \times 10^{-3}$$

$$= 26.297 \text{ kN}$$

Maximum allowable shear stress

22.6.6

$$v_{cu} = v_c + \phi \cdot 0.66 \cdot \sqrt{f_{c'}}$$

$$= .66407 + .75 \times 0.66 \times \sqrt{30}$$

$$= 3.375 \text{ MPa}$$

Actual Shear stress:

$$v = \frac{V \cdot 1 \times 10^3}{A_{sh}}$$

$$= \frac{8.308 \times 1 \times 10^3}{39600}$$

$$= 0.2098 \text{ MPa}$$

Minimum links 9.6.3.3

$$A_{sv/svn} = \frac{0.35 \cdot b}{f_{yv}}$$
$$= \frac{0.35 \times 220}{250}$$
$$= 0.3080$$

$$v_1 = \sqrt{f_{c'}} = 5.477 \text{ MPa}$$

$$A_{sv/svn1} = \frac{0.062 \cdot b \cdot v_1}{f_{yv}}$$
$$= \frac{0.062 \times 220 \times 5.4772}{250}$$
$$= 0.2988$$

9.6.3.3

$$A_{svn} = \max(A_{svn}, A_{svn1})$$
$$= \max(, .29884)$$
$$= 0.2988$$